Synthetic Handbook — LV Induction Motor Applications (DEMO-MT-004)

SYNTHETIC DEMONSTRATION DOCUMENT — fictional data, not a real standard

Fictional handbook for 400 V three-phase squirrel-cage motors used to exercise rating lookups, conversions and cable/protector sizing.

Chapter 1 — Motor DEMO-M-203 (315 kW pump drive)

Quantity	Value	Condition
Rated output	315 kW	shaft power, continuous S1
Rated apparent power	345 kVA	at rated load
Rated voltage	400 V	phase-to-phase, 50 Hz
Voltage tolerance	–10 % to +5 %	at the terminals
Rated current	515 A	calculated from 345 kVA
Full-load efficiency	96.3 %	at 75 °C winding
Power factor	0.912	at rated load, typical
Locked-rotor current	6.2 × I_rated	= 3.2 kA (rounded)
Starting torque	1.9 × T_rated	DOL start
Breakdown torque	2.6 × T_rated	typical
Insulation class / rise	F / B	class F, rise limited to B
Maximum ambient	40 °C	at altitude ≤ 1000 m
Frame size	IEC 355M	IC 411 fan-cooled

Cross-check (authoritative for verification): rated current I = 345 kVA / (√3 × 400 V) = 498 A — the tabulated 515 A includes the ±5 % voltage tolerance contingency of 3.4 % and measurement rounding; use the tabulated value for protection settings and the 498 A value for cable sizing references.

Chapter 2 — Starting and protection

2.1 DOL starting

The maximum DOL starting current is 3.2 kA with a peak of 14 kA at rated voltage. A minimum bus voltage of 336 V (84 % of 400 V) is required at the MCC during the 6 s acceleration time.

2.2 Fuse and contactor ratings

Item	Rating
AC3 duty contactor	560 A
Fuse (aM) for short-circuit protection	800 A
Overload relay setting range	450–560 A
Earthing conductor for the motor body	25 mm² copper

2.3 VSD operation

With a voltage-source inverter the motor shall be fed via 10 m of symmetric shielded cable; V/f control keeps 8:1 speed range at constant torque. The rated inverter output is 345 kVA at 400 V. Output frequency range 0.5 to 100 Hz. Bearing current mitigation: insulated NDE bearing plus 50 mm² shaft grounding brush.

Chapter 3 — Maintenance limits

Vibration velocity alarm at 7.1 mm/s RMS (speed 1485 rpm); trip at 11.0 mm/s. Winding resistance imbalance shall not exceed 2 %. Insulation resistance at 40 °C with 1000 V DC: minimum 100 MΩ; temperature-corrected value uses the handbook rule of factor 2 per −10 °C.

SAFETY: rotating machine — isolation, blocking of the pump coupling and LOTO apply before any terminal work.